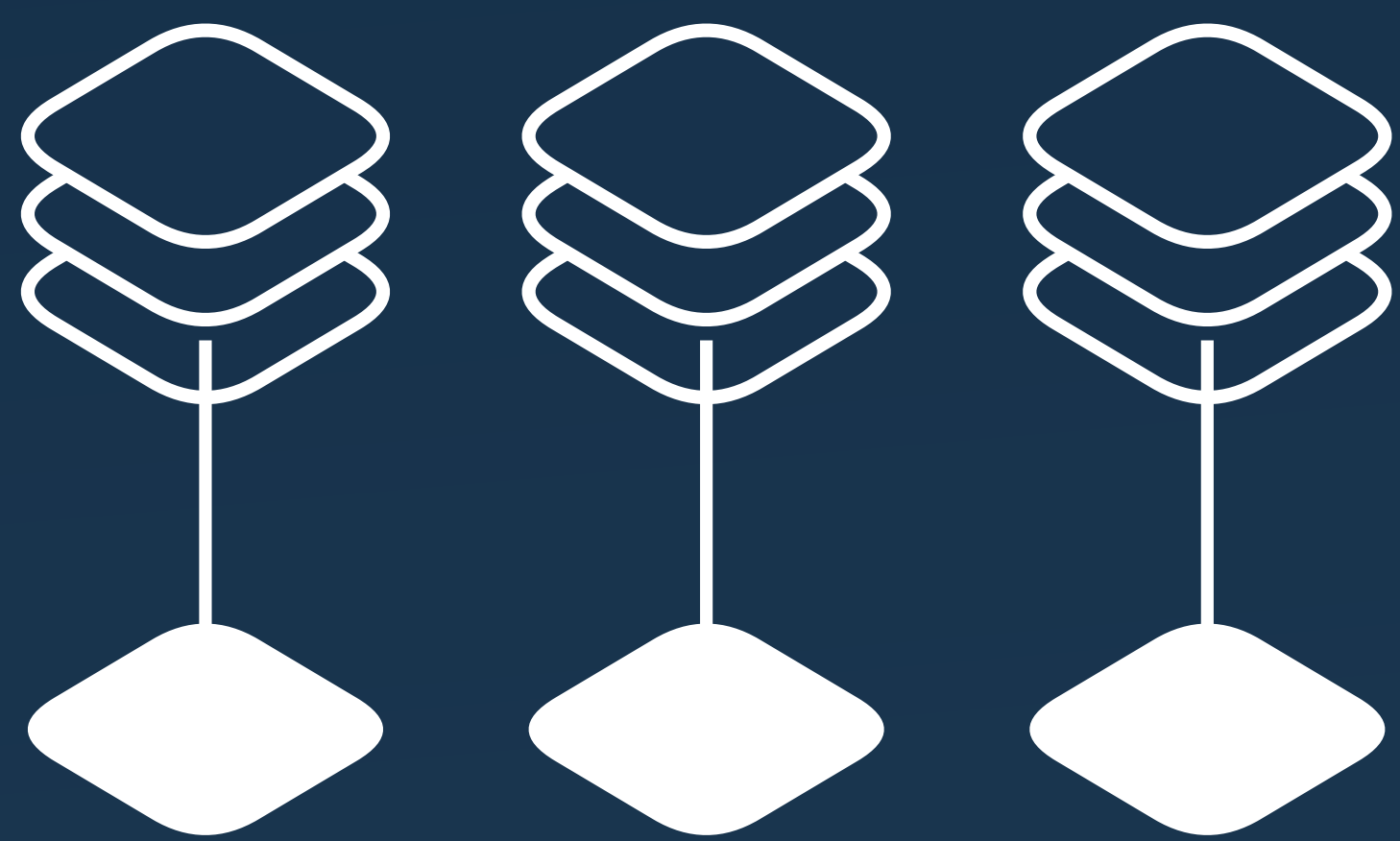




Which layer of a frozen vision foundation model should you probe?

None alone.

For dense prediction, try **AMDF**: it fuses **all** layers, for **every** patch.



On DINOv3-L, NYUv2 probes **peak at different depths**: normals $\approx$ L18, depth $\approx$ L21, segmentation L24.

$\approx 22\times$

fewer trainable parameters than concatenating all layers ($\approx 1.2\text{M}$ vs 26.2M , DINOv3-L, dense tasks), at the **2nd-best** mean rank.

5 / 5

benchmarks where AMDF has a **better** mean rank than last-layer, mean, weighted and per-position fusion; 1st on depth.

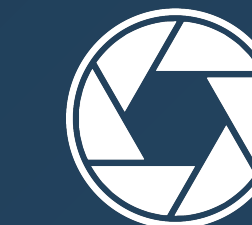
Frozen backbone · no layer search
one probe, any head
Full tables → take a **handout**

AMDF: Holistic Probing of Vision Foundation Models with Adaptive Multi-Layer Dense Fusion

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Different tasks live at different depths

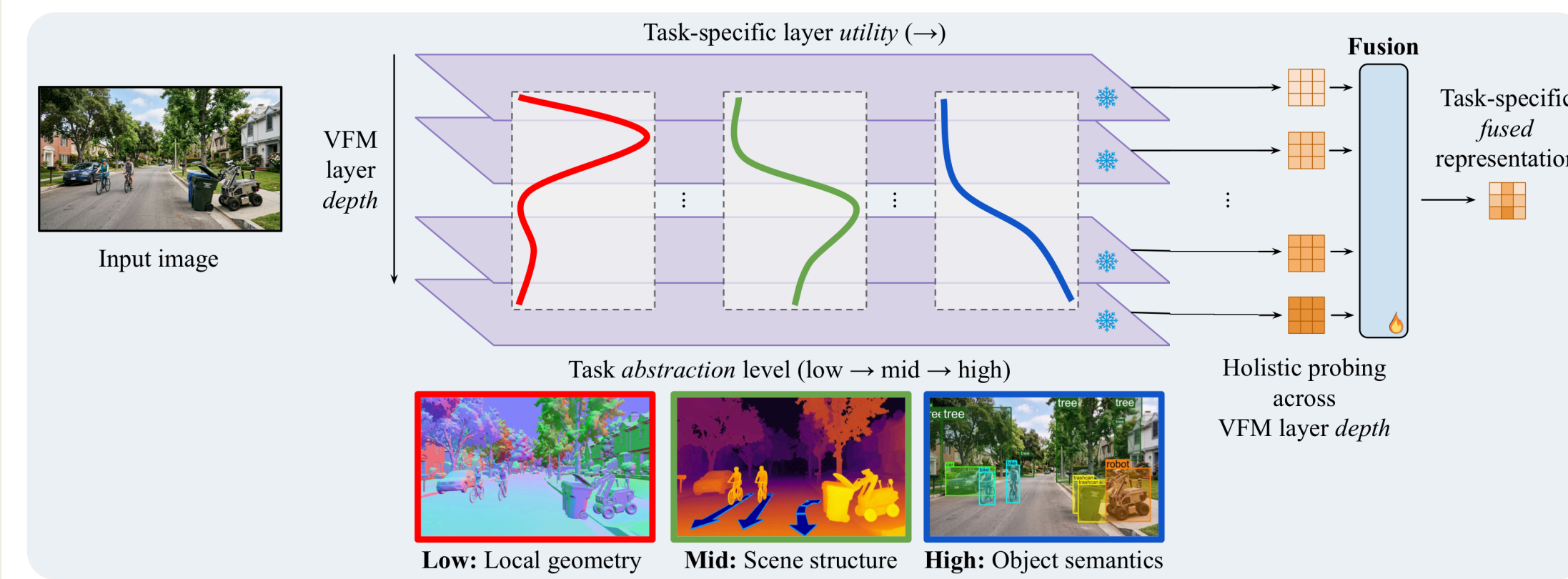


Fig. 1: Geometry peaks in middle layers, semantics near the end. AMDF reads every frozen layer and fuses them per patch.

Problem #1: the last layer is not the best

X Probing only the final layer misses what a task needs.

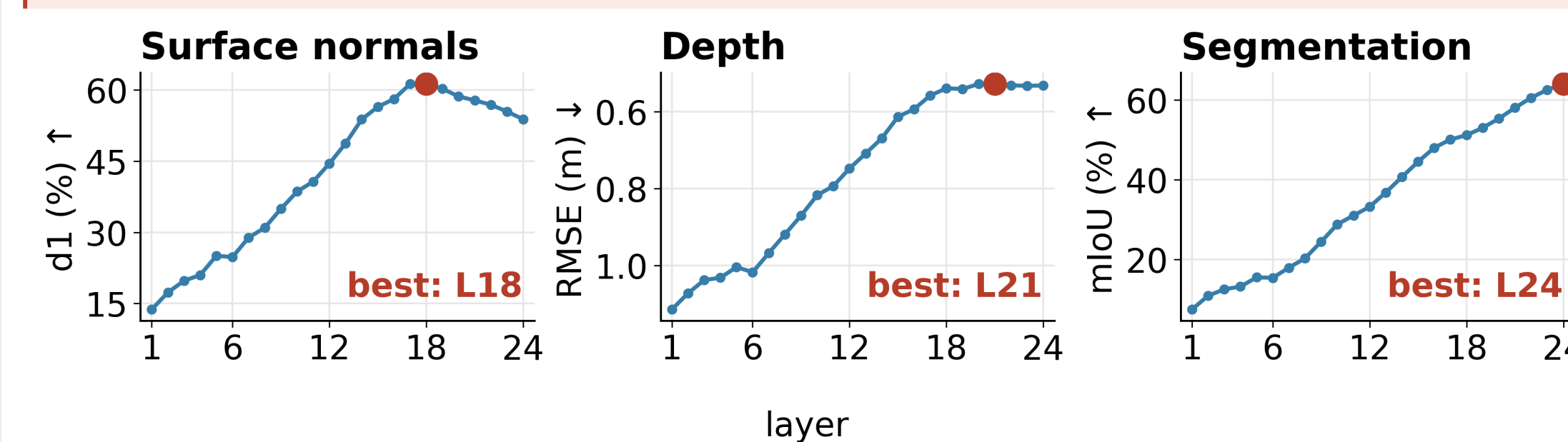


Fig. 2: One probe per layer of a frozen DINOv3-L on NYUv2 (single seed): the best layer moves deeper as the task gets more semantic, peaking around L18 → L21 → L24.

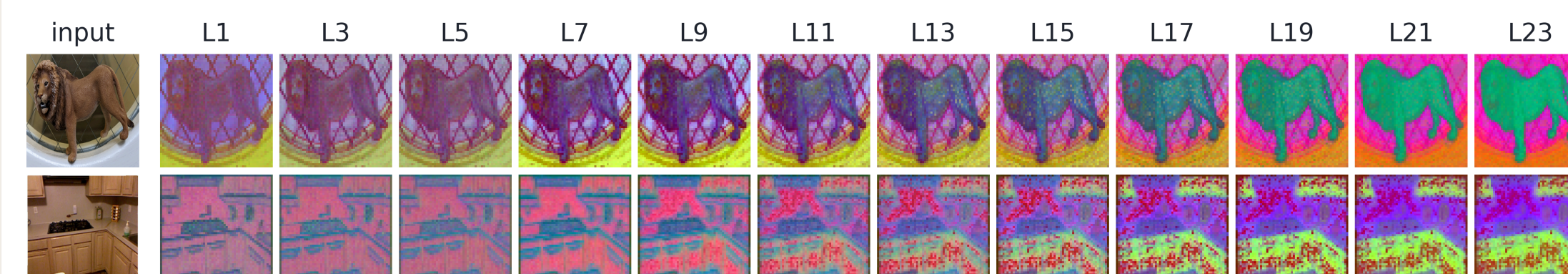


Fig. 3: Frozen PESpatial-L patch features (one PCA per image, 770 px input): objects separate from the background only in deep layers.

Problem #2: using all layers is costly

X Concatenating all layers is accurate but heavy: **26.2M** trainable params vs **1.17M** for AMDF (DINOv3-L, normals, incl. head).

✓ AMDF learns per patch how much of each layer to use.

Setup

7 backbones × 5 benchmarks

DINOv3-S/B/L · MAE-L · SigLIP2-L · V-JEPA2.1-L · PESpatial-L — NYUv2 normals, depth (log head), seg. · Taskonomy · NAVI

Baselines

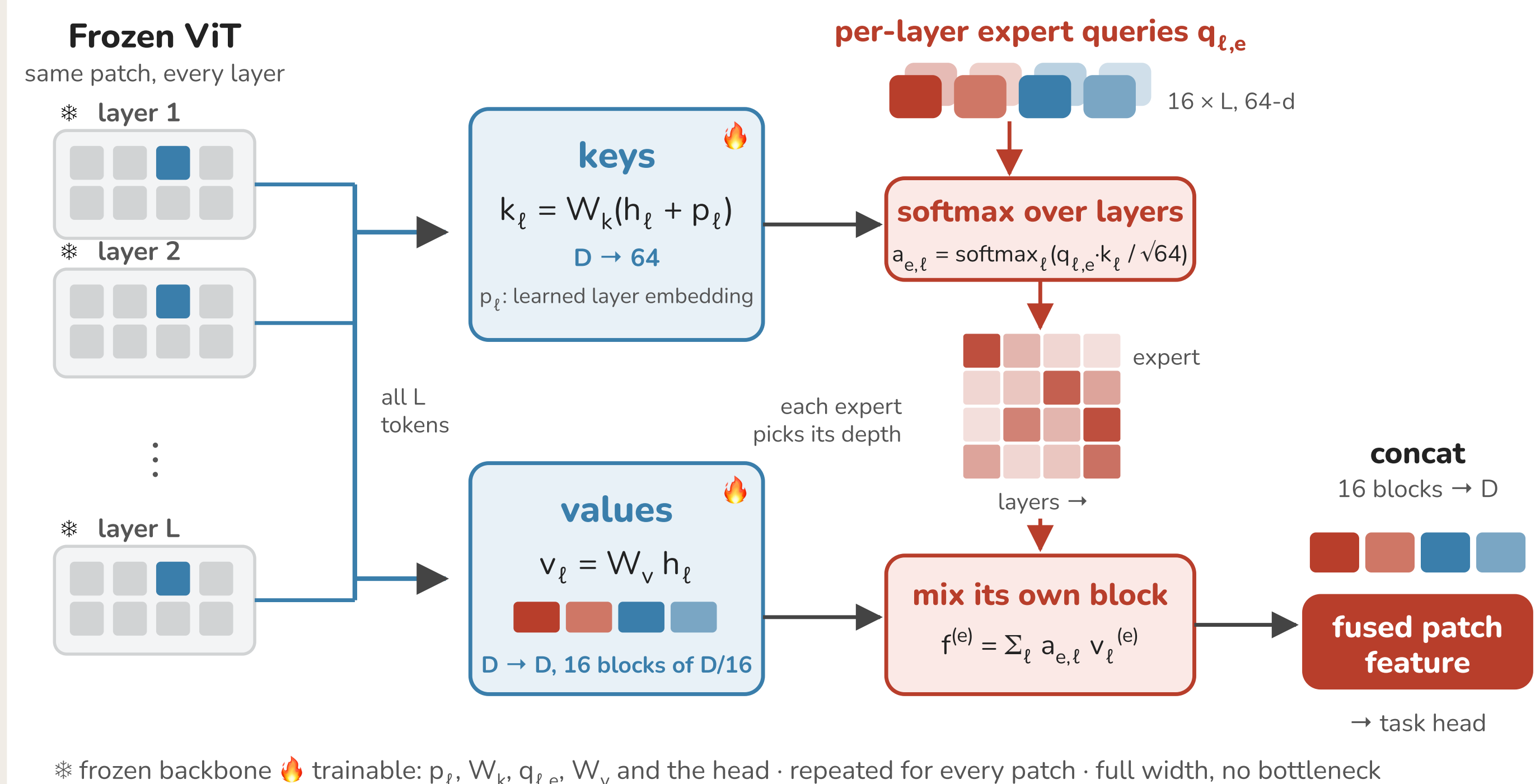
Last · Avg · Weighted (one layer mix) · SpatialWeights (per-position mix) · Concat (1×1 proj. + GELU)

Protocol

full width, 64-d keys/queries · same recipe per table · mean of ≤ 3 seeds (few cells 1–2) · best val. epoch

Solution: AMDF — depth-aware fusion

For every patch, 16 experts — with their own query for every layer — score all layers with 64-d keys, take a **softmax over layers** and mix their own block of full-width channels, so each expert picks its own depth.



⚡ frozen backbone 🔥 trainable: $p_l, W_k, q_{l,e}, W_v$ and the head · repeated for every patch · full width, no bottleneck

One probe, any head: the fused features feed a linear dense head (1×1 conv, then bilinear upsampling: normals, depth, segmentation, Taskonomy) or a **pairwise** head (pool both images, concatenate, MLP → camera pose). The backbone stays frozen; only the probe and head train.

Its attention follows layer utility

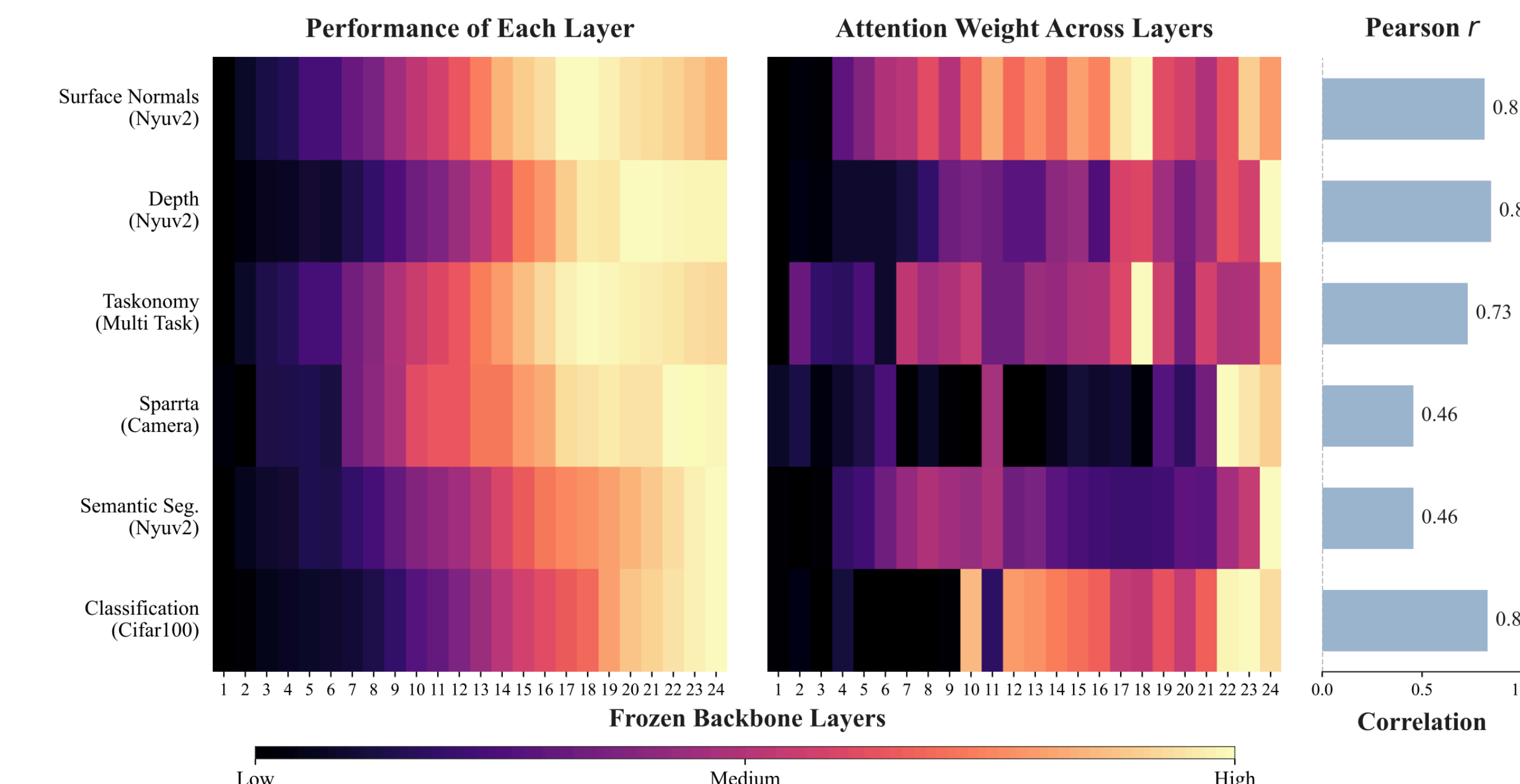


Fig. 4: Measured on an earlier AMDF variant (with per-layer refinement). AMDF's learned depth-attention (middle) tracks per-layer probe performance (left) on DINOv3-L/16. Alignment (Pearson r , right) is strongest for dense geometric tasks (depth $r=0.85$, normals $r=0.81$, Taskonomy $r=0.73$) and classification ($r=0.83$), weaker for camera and segmentation ($r=0.46$). Datasets differ from the tables (e.g. CIFAR-100, SparrTa).

Ablation: per-layer vs. shared queries

Normals	Taskonomy	Segm.	Depth	Pose
+1.62 d1	−0.73	+0.16 mIoU	+0.51 d1	−2.37° rot.
7/7 backbones	L1×100	6/7 backbones	5/7 backbones	5/7 backbones
	7/7 backbones			

Paired gains (same seeds), mean over 7 backbones; per-layer queries cost +23.5k parameters ($\approx 2\%$). Pose gain is mostly SigLIP2-L ($\sim 13^\circ$).

Results

$\approx 22\times$
fewer params
than Concat

2nd
mean rank
(1st on depth)

5×7
benchmarks ×
backbones

Mean rank per benchmark (1 = best, ★ = best in table)

	Normals	Depth	Segm.	Taskonomy	Pose
AMDF (ours)	2.00	1.43 ★	1.86	2.00	1.93
ConcatLayers	1.00 ★	1.71	1.57 ★	1.00 ★	1.21 ★
WeightedLayers	4.14	4.00	4.14	3.29	3.89
SpatialWeights	3.14	4.14	4.43	4.00	4.00
LastLayer	5.43	4.14	3.86	5.29	4.96
AvgLayer	5.29	5.57	5.14	5.43	5.00

Fig. 5: Each cell is a method's rank averaged over the 7 backbones of that table (NAVI: rotation and translation). ConcatLayers ranks first on 4 of 5 benchmarks, AMDF on depth; AMDF has a better mean rank than the four cheap baselines in every table. Full tables: [handout](#).

Accuracy vs. trainable parameters

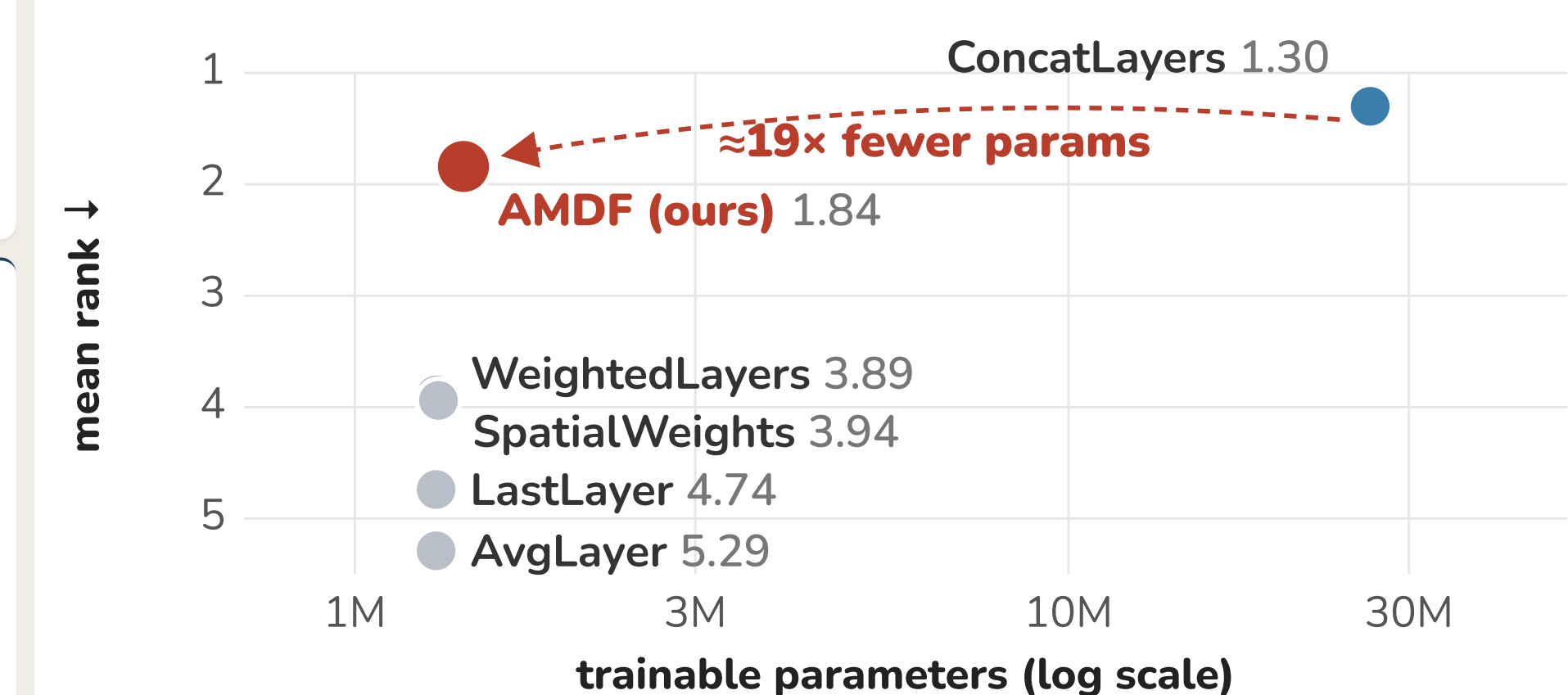


Fig. 6: Mean of the 5 per-table ranks vs. trainable parameters (probe + head, DINOv3-L, averaged over the same tasks; $\approx 19\times$ here, $\approx 22\times$ on dense tasks alone).

Take-home

- Useful features are **spread across depth**, and the best depth depends on the task.
- AMDF beats all four cheap fusions by mean rank in every table and is second only to ConcatLayers (1st on depth), with **$\approx 22\times$ fewer params**.
- A fixed per-position mix (SpatialWeights) is no better than one global mix (mean rank 3.94 vs 3.89): choosing layers **from the patch's content** is what helps.
- Per-layer expert queries** improve the mean on all 5 benchmarks (on every backbone for normals and Taskonomy) for +2% parameters.
- Study frozen VFMs as **full hierarchies**.